SHIVANI KUMARI

+886 0958385873

shivanikumaribit1998@gmail.com ⋄ LinkedIn • GitHub

OBJECTIVE

PhD researcher in AI and biomedical imaging with practical experience in OCTA, MRI, CT, and ultrasound-based signal processing. Passionate about advancing medical image analysis, multimodal data fusion, and deep learning for healthcare applications. Eager to contribute to impactful research in interdisciplinary and collaborative environments.

EDUCATION

Doctor of Philosophy (PhD) in Data Science and AI Applications, National Yunlin University of Science and Technology, Taiwan *2024 – Present*

Relevant Coursework: Artificial Intelligence, Internet of Things (IoT), Machine Learning, Deep Learning, Natural Language Processing, Computer Vision

Research Focus: AI-driven non-invasive blood glucose monitoring using ultrasound, OCTA, MRI, and CT imaging modalities; emphasis on multimodal signal processing and medical image analysis

Master's in Sciences, Computer Science and Information Engineering, YunTech, Taiwan 2021 - 2023

Relevant Coursework: Machine Learning, Programming, DBMS, Signal Processing, Image Processing, Linux, C++, Python, Biometric Authentication.

Bachelor of Technology, Birla Institute of Technology, Mesra, India 2017 - 2021

Relevant Coursework: VLSI, MATLAB, Networking, Mobile Networking, TCP/IP, Semiconductors, IoT, Routers, WiFi, Microcontrollers, DBMS.

SKILLS

Medical Imaging	OCTA, MRI, CT, Ultrasound, Radiomics, Photoacoustic Imaging
Machine Learning	Deep Learning, Supervised/Unsupervised Learning, Computer Vision, Federated Learni
Programming Languages	Python, C/C++, SQL, HTML, CSS, JavaScript, Go
Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV
Developer Tools	Google Colab, Anaconda, VS Code, MATLAB, Git, Postman, Spyder, Blender
Systems	Linux, Google Cloud Platform, SDLC
Soft Skills	Scientific Writing, Team Collaboration, Communication, Critical Thinking
Languages	English, Hindi, Bengali, Elementary Chinese
Interests	Yoga, Dance, Basketball, Badminton, Reading, Writing

EXPERIENCE

Research Intern, National Dong Hwa University (NDHU), Hualien, Taiwan Jul 2025 – Aug 2025

Student Laboratory Exchange Program

- Conducted AI-driven biomedical research focused on non-invasive healthcare technologies under the guidance of Assoc. Prof. Wei-Che Chien
- Collaborated in a cross-institutional setting as part of the National University System of Taiwan
- Strengthened expertise in Data Science AI applications for healthcare

Certificate issued by National Yunlin University of Science and Technology (YunTech)

Research Assistant, National Yunlin University of Science and Technology, Taiwan Feb 2024 – Present

Project: Application of Non-Invasive Blood Glucose Level Sensor Glucometer Using Ultrasound in Monitoring Blood Glucose

- Developed real-time AI models using OCTA, ultrasound, CT data for glucose prediction
- Experience with SwinUNet, U-Net, radiomics, and statistical modeling
- Collaborated with medical institutions for data collection and clinical integration

Software Developer, Cefinity: Hannlync Inc., Hsinchu, Taiwan Aug 2023 - Dec 2023

Full-stack development, Data Analysis, API design

Intern, TATA Steel Limited, Jamshedpur, India June 2020 - Aug 2020
Project: Active Time Prediction Model: Data prediction using Regression, Neural Networks.
Intern, GIMER Medical (Neurostimulation), Taipei, Taiwan May 2019 - June 2019
Project: Using Arms Embedded System to Develop Tens Application.
research assistant, Yuntech, Taiwan February 2024 - present
Project: Application of Non-Invasive Blood Glucose Level Sensor Glucometer Using Ultrasound in Monitoring Blood Glucose

PROJECTS

Application of Non-Invasive Blood Glucose Level Sensor Glucometer Using Ultrasound in Monitoring Blood Glucose: Real-time project in collaboration with Hualien Tzu Chi Hospital, Buddhist Tzu Chi Medical Foundation, Hualien, Taiwan. Used *IOT SYSTEM DESIGN, MATLAB and Python* for signal processing and machine learning analysis to improve glucose measurement accuracy.
Vin Big Data Chest X-ray Abnormalities Detection: Developed an automated system using *TensorFlow and OpenCV* to detect abnormalities in chest radiographs. Implemented deep learning techniques to aid in faster diagnosis.
RSNA-MICCAI Brain Tumor Classification: Utilized *Scikit-learn and PyTorch* to classify MRI scans for predicting genetic biomarkers, improving early-stage detection.
IoT-Based Smart Gardening: Designed an automated irrigation system using *Arduino and IoT sensors.* Integrated real-time data analytics and ML models to optimize water usage based on environmental conditions.
Employee Clock-in System: Developed a clock-in system using *Django and PostgreSQL*, incorporating biometric authentication for enhanced security.

COURSES

Machine Learning — Stanford University (Coursera)
IoT and Embedded Systems — University of California, Irvine (Coursera)
The Data Scientist's Toolbox — Johns Hopkins University (Coursera)
Data Visualization with Python — IBM Skills Network
DB and SQL for Data Science — IBM Skills Network
Data Science Bootcamp 2021 — Udemy
AI For All — Government of India, MeitY (National AI Portal)

AWARDS AND CERTIFICATES

Best Paper Runner Up Award — IEEE SmartIoT 2024, Shenzhen, China
Paper: UAV-Based Deep Learning with Tiny-YOLOv9 for Revolutionizing Paddy Rice Disease Detection
MOE Scholarship (Taiwan): Full scholarship for master's program.
TATA Steel Scholar (India): Full scholarship for bachelor's program.
VinBigData Chest X-ray Detection: 1st Prize.
Yoga: Certified Yoga Practitioner (Alliance USA).

PUBLICATIONS

Shivani Kumari, et al. *A Comparative Study on ECG Biometric Authentication using Machine Learning.*
National Conference on Web Intelligence and Applications (NCWIA), 2022.

Shivani Kumari, et al. *Research on a Biometric Key Scheme for Identity Recognition Based on Concise PPG Signal Pattern.*
National Conference on Web Intelligence and Applications (NCWIA), 2023.

Shivani Kumari, et al. *UAV-Based Deep Learning with Tiny-YOLOv9 for Revolutionizing Paddy Rice Disease Detection.*
The 8th IEEE International Conference on Smart Internet of Things (SmartIoT), November 2024, Shenzhen, China.
Awarded: *Best Paper Runner-Up Award*

Shivani Kumari, et al. *Blood Glucose Alarm System using K-Nearest Neighbors.*
International Workshop on Advanced Image Technology (IWAIT), January 2025, Yunlin, Taiwan.

Shivani Kumari, et al. *Non-Invasive Glucose Monitoring Using Ultrasound and AI: A Comprehensive Review.*
Journal of Innovative Technology (JIT), Accepted 2025.

Shivani Kumari, Arun Kumar Sangaiah, Wen-Fong Wang, Honda Hsu, Chung-Chian Hsu, and P. Balakrishnan. *Simulation-Driven Design of a Non-Invasive Photoacoustic Glucose Monitoring Device With IoT Integration and Hybrid Energy Harvesting.*
IEEE Internet of Things Magazine, Early Access, pp. 1–10, Dec. 29, 2025.
DOI: 10.1109/MIOT.2025.3639278

Shivani Kumari, Arun Kumar Sangaiah, and Wen-Fong Wang. *Multi-scale vessel and lesion feature fusion for early diabetic retinopathy detection using fundus images.*
Proceedings of SPIE, Vol. 14072, International Workshop on Advanced Imaging Technology (IWAIT 2026), Paper 140720A, February 2026, Kaohsiung, Taiwan.
DOI: 10.1117/12.3109894

Under Review / In Preparation

Shivani Kumari, et al. *Molecularly Informed Artificial Pancreas Using Single-Cell Transcriptomics.*
Manuscript in final preparation.

Shivani Kumari, et al. *LLM-Enhanced OCTA-Based Glucose Monitoring: A Multimodal Framework.*
Currently under modification.

Shivani Kumari, Arun Kumar Sangaiah, and Wen-Fong Wang. *Physiology-Aligned Non-Invasive Glucose Prediction Using Tabular Foundation Models and Stacked Ensembles under Leave-One-Subject-Out Validation.*
Manuscript in preparation.

Shivani Kumari, Arun Kumar Sangaiah, and Wen-Fong Wang. *Integrating Anatomical and Pathological Priors for Diabetic Retinopathy Grading: A Systematic Study.*
Manuscript in preparation.